

Toward explaining typologically rare strengthening: evidence from Loushao Xiang Chinese

The Xiang dialects is one of the major dialect groups of Chinese, located in Hunan province in southern-central China. The Loushao subgroup of Xiang shows interesting sound changes where some previous alveolopalatal affricates become coronal or velar stops, referred to as hardening and velarization respectively. For example in Lianyuan, 中 *zhōng* ‘middle’ Proto-Loushao (PLS) **tejɔŋ* > *ʈaŋ*, 展 *zhǎn* ‘unfurl’ PLS **tejən* > *ki* (tones omitted). Hardening and velarization are both strengthenings where previous affricates change to stops, which are very rare cross-linguistically and the trigger and mechanism remain unclear (Bybee & Easterday 2019).

Several difficulties are involved in explaining these typologically rare sound changes. First, comparative evidence confirms that these sounds go back to coronal affricates and not stops. Furthermore, the hardening and velarization patterns are highly dialect dependent. Although a high front element after the onset seems to be the shared environment, the dialects show different hardening/velarization results with different places of articulation and may or may not show hardening/velarization depending on following roundedness. The sound change therefore cannot be reconstructed back to the proto stage, but must be explained through individual changes. The recurrence of such a rare change across multiple dialects is therefore unexpected. Second, although there are chain shifts such as *t, t^h → ts, ts^h → Ø, h* found in some other Chinese dialects, no such chain shifts are involved in the Loushao dialects. They therefore cannot be explained by chain shifts in Loushao. Third, this is a depalatalization change that is the opposite of the cross-linguistically common palatalization process. The rarity of the sound change posits challenges on plausible explanations on the trigger and mechanism.

My current study examines the patterns of hardening and velarization among eleven Loushao dialects. The dataset includes eight dialects from Coblin (2011) and three additional dialects described in later fieldworks: Hengshan Houshan by Yang (2011), Wugang by Zeng (2023), and Suining by Li (2019). A modified Swadesh 200-wordlist with about 250 cognates is used as the basis for reconstructing PLS, with adjustments aiming to include more native cognate words and to cover the full phonological inventory.

I explore the hypothesis of taking in vowel apicalization together to explain the typologically rare sound changes. Vowel apicalization is a process of high vowel raising attested cross-linguistically in the Xiang dialects as well as the Chinese dialects and the Bantho languages in Africa (Faytak 2014). In the Loushao dialects, vowel apicalization is a common reflex for PLS syllables with alveolopalatal affricates and a high front main vowel, i.e. changes such as PLS **tei* > *tɕʰ/ʈɕʰ*. Previous studies have treated hardening/velarization and vowel apicalization as independent developments (Coblin 2011). I propose that both vowel apicalization and hardening/velarization are triggered by crowded high front vowel space caused by vowel raising, while the continued raising shift shows different effect on different syllable structures. This hypothesis offers a unified account of hardening, velarization, and vowel apicalization while accommodating the dialect-specific reflexes, and provides a new perspective on the interaction between consonantal and vocalic strengthening.